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FACULTY OF ENGINEERING AND TECHNOLOGY

Group 11 WAR

OPTIMIZATION ASSIGNMENT

Group members

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QUESTION ONE: OPTIMISATION OF A WATER DISTRIBUTION PIPELINE

A water supply authority plans to extend its distribution network to serve two residential zones, A and B. Water will be conveyed from an existing treatment reservoir through two pumping and distribution branches. The authority must determine the daily quantities supplied to the two zones while minimising the operating cost. The quantities are represented by x and y, respectively, measured in thousands of cubic metres per day.

The estimated operating cost is represented by the nonlinear objective function:

$$\textbf{Minimise} \quad Z = 4x + 6y + 0.5x^2 + 0.3y^2$$

The authority specifies the following requirements:

- The combined daily supply to the two zones must be exactly 20,000 m^3.

- Zone A must receive at least 5,000 m³ per day.
- Zone B must receive at least 4,000 m³ per day.
- The combined pumping capacity must not exceed 30,000 m³ per day.
- The hydraulic operating condition must satisfy the nonlinear limitation stated below.

The mathematical constraints are:

$$\mathbf{x + y = 20}$$

$$\mathbf{x \geq 5; \quad y \geq 4; \quad x + y \leq 30}$$

$$\mathbf{x^2 + y^2 \leq 400}$$

Assume $x \geq 0$ and $y \geq 0$.

Required:

a) Write the complete mathematical optimisation model and develop a MATLAB programme using fmincon to obtain the optimal allocation.

b) Output plots for water distribution cost optimisation

```
%matlab code
% v(1)=x;
% v(2)=y;
%estimated operating cost
Z=@(v) (4.*v(1)+6.*v(2)+0.5.*v(1).^2+0.3.*v(2).^2);
x0=[10,10];
A=[1,1];
b=30;
Aeq=[1,1];
beq=20;
lb=[4;5];
ub=[];
nonlcon=@(v)deal(v(1).^2+v(2).^2-400,[]);
[v,Zmin]=fmincon(Z,x0,A,b,Aeq,beq,lb,ub,nonlcon);
fprintf('Zone A= %.4fthousand m3/day\n',v(1));
fprintf('Zone B= %.4fthousand m3/day\n',v(2));
fprintf('Minimum cost= % .2fM\n',Zmin);
% objective counter plot
[X, Y]=meshgrid(5:0.1:16,0.1:15);
Z=4.*X+6.*Y+0.5.*X.^2+0.3.*Y.^2;
figure;
contour(X,Y,Z,25);
```

```
hold on
plot(v(1),v(2));
xlabel('Zone A supply(thousand m3/day');
ylabel('Zone B supply(thousand m3/day');
title('Water Distribution cost optimization');
grid on;
```

QUESTION TWO: OPTIMISATION OF A REINFORCED CONCRETE BEAM

A civil engineering consultancy is designing a simply supported reinforced concrete beam for a commercial building. The beam has a span of 6 m and carries a uniformly distributed load of 20 kN/m. The design team wishes to minimise the concrete volume while ensuring that the beam satisfies a preliminary bending-capacity requirement and practical dimensional limitations.

Let b be the beam width (m) and d be the overall beam depth (m). The concrete volume is:

$$\textbf{Minimise} \quad V = 6bd$$

The beam's maximum bending moment is calculated using:

$$M_{max} = wL^2 / 8$$

For a rectangular section, the elastic section modulus is $bd^2/6$. For this preliminary exercise, require the section's bending capacity to equal the specified design requirement:

$$bd^2 = 0.054$$

The design must also satisfy the following restrictions:

$$0.20 \leq b \leq 0.40$$

$$0.30 \leq d \leq 0.70$$

$$b + d \leq 1.00$$

$$bd \geq 0.12$$

The equality $bd^2 = 0.054$ represents the required preliminary bending-capacity condition. The product constraint $bd \geq 0.12$ is a minimum cross-sectional area requirement.

Required:

a) Develop a MATLAB programme using ga to determine the optimum beam dimensions. State the initial guess used.

b) Present and interpret the optimal beam dimensions and minimum concrete volume.

```
%matlab code
% x(1)=b;
% x(2)=d;
nvars=2;
V=@(x) (6.*x(1).*x(2));
```

```
A=[1,1];
b=1.00;
Aeq=[];
beq=[];
lb=[0.20,0.3];
ub=[0.4,0.7];
nonlcon=@(x) deal(0.12-x(1).*x(2),x(1).*x(2).^2-0.054);
[x,Vmin]=ga(V,nvars,A,b,Aeq,beq,lb,ub,nonlcon);



fprintf('Optimal width b= %.4fm\n',x(1));
fprintf('Optimal width d= %.4fm\n',x(2));
fprintf('Minimum volume= %.4fm3\n',Vmin);

[B,D]=meshgrid(0.2:0.005:0.4,0.3:0.005:0.7);
V=6.*B.*D;
figure;
contour(B,D,V,25);
hold on

plot(x(1),x(2),MarkerSize=12);
xlabel('Beam width b (m)');
ylabel('Beam depth D in (m)');
title('Concrete volume optimisation');
grid on
```

QUESTION THREE: OPTIMISATION OF A SOLAR-POWERED WATER PUMPING SYSTEM

A rural water supply authority intends to install a solar-powered pumping system to supply a community. The engineering team must select the number of solar panels and the daily pumping duration while minimising the simplified system cost and meeting the required daily water production.

Let x represent the number of solar panels and y represent the daily pumping duration in hours. For this exercise, the daily water production model is Q = 4xy, measured in cubic metres per day. The simplified cost function, in thousands of Uganda shillings, is:

$$\textbf{Minimise} \quad Z = 500x + 200y + 10x^2 + 5y^2$$

The system must satisfy the following requirements:

$$4xy = 100$$

$$x \geq 5; \quad y \geq 5; \quad x + y \leq 30$$

$$x^2 + 2y^2 \leq 500$$

The water-production equality requires the system to produce exactly 100 m^3/day for the stated design demand. The nonlinear energy limitation is a simplified operating constraint. Assume x is a positive integer because solar panels are installed as whole units, while y is continuous.

f) Develop a MATLAB solution for the problem.

g) Report the optimal number of panels, pumping duration and minimum cost.

```
%matlab code
% c(1)=x;
% c(2)=y;
nvars=2;
P=@(c) (500.*c(1)+200.*c(2)+10.*c(1).^2+5.*c(2));
A=[1,1];
b=30;
Aeq=[];
beq=[];
lb=[5,5];
ub=[];
nonlcon=@(c) deal(c(1).^2+c(2).^2-500,4.*c(1).*c(2).^2-100);
[c,Pmin]=ga(P,nvars,A,b,Aeq,beq,lb,ub,nonlcon);



fprintf('Optimal panels = %d\n',c(1));
fprintf('Pumping duration= %.4f hours/day\n',c(2));
fprintf('Minimum cost= UGX  %.2f\n',Pmin);

[X,Y]=meshgrid(0.2:0.005:0.4,0.3:0.005:0.7);
Z= 500.*X + 200.*Y + 10.*X.^2+ 5.*Y.^2;
figure;
contour(X,Y,Z,25);
hold on

plot(c(1),c(2),MarkerSize=12);
xlabel('optimal panels');
```

```
ylabel('pumping duration(hours/day)');
title('Minimum cost optimization');
grid on
```